



ESSE ALTERNATE ASSESSMENT

Topic: Energy Inefficient Buildings



BY: MELWIN MATHEW
S3 CS GAMMA
U2503285

ABSTRACT

Buildings consume large amounts of electricity for air-conditioning, lighting, computers, laboratory equipment and other services. While this energy consumption is necessary for normal building operation, a considerable amount can be unnecessarily consumed when electrical equipment operates beyond required working hours or when rooms remain powered despite having little or no usage. Identifying such wastage is therefore an important step towards improving building energy efficiency.

This report proposes an **Energy Waste Detection and Audit Portal**, a software-based web application designed to identify potential energy wastage by analysing building schedules, equipment operating periods and energy-consumption data. Instead of simply displaying total energy consumption, the system focuses on identifying **where and when energy may be unnecessarily consumed**.

The portal compares scheduled room usage with actual equipment operation and generates alerts when equipment operates beyond expected periods. It can also estimate potentially avoidable energy consumption and provide room-level energy audit information. An interactive "What-If" simulator allows users to estimate potential savings from reducing unnecessary equipment operation.

The proposed system can be implemented as a software-only prototype using HTML, CSS and JavaScript with simulated or manually entered data. It can later be integrated with smart meters, IoT devices and building management systems. The solution supports SDG 7 by promoting energy efficiency and also contributes to SDG 11 and SDG 12 by encouraging sustainable building operation and responsible resource consumption.

Keywords: Energy Efficiency, Energy Waste, Energy Audit, Green Buildings, Sustainable Buildings, Web Application, Energy Management

INTRODUCTION

Buildings form an important part of modern infrastructure and support activities such as education, business, healthcare and public services. Electricity is required to operate lighting, air-conditioning, computers, laboratory equipment, ventilation systems and other building services. As the number and complexity of buildings increase, improving their energy efficiency becomes increasingly important for both environmental sustainability and operational cost reduction.

Energy inefficiency does not necessarily mean that a building uses inefficient equipment. Energy can also be wasted because equipment is operated unnecessarily. For example, an air-conditioner may remain switched on after a class has ended, lights may remain active in an unused room, or computers in a laboratory may continue operating after working hours.

These individual incidents may appear insignificant, but repeated wastage across many rooms and buildings can result in substantial energy consumption.

Traditional energy-monitoring systems generally focus on reporting consumption. A dashboard may show that a building consumed 1,000 kWh in a particular period, but this information alone does not explain where the energy was unnecessarily consumed or what caused the increase.

Therefore, there is a need for a system that can convert energy-use information into **actionable information about potential energy wastage**.

The proposed Energy Waste Detection and Audit Portal addresses this issue through a software-based approach. The system analyses building operating information, identifies deviations from expected usage and presents them as energy-waste events that can be investigated by building administrators.

The basic principle of the proposed system is:

Collect → Compare → Detect → Quantify → Improve

PROBLEM STATEMENT

The assigned environmental problem is **Energy-Inefficient Buildings**.

Buildings may consume unnecessary energy due to inefficient operating practices such as:

- air-conditioning operating after scheduled usage;
- lights remaining switched on in unused rooms;
- computers and laboratory equipment being left running;
- equipment operating outside normal building hours;
- excessive equipment operating duration;
- lack of detailed monitoring of room-level energy usage.

A conventional electricity bill or basic energy dashboard provides information about total consumption but generally does not identify individual energy-waste events.

For example, consider a classroom scheduled from 9:00 AM to 10:00 AM. If the air-conditioner continues operating until 12:00 PM, the building has potentially consumed additional electricity without corresponding scheduled usage.

The problem can therefore be stated as:

How can a software-based system identify potential energy wastage in buildings by comparing expected building operation with actual equipment usage and quantify the resulting energy-saving opportunities?

SUSTAINABILITY ANALYSIS

Sustainable Development Goal 7

The primary SDG associated with the proposed solution is **SDG 7: Affordable and Clean Energy**.

SDG 7 promotes access to affordable, reliable and sustainable energy and includes improving energy efficiency as an important target. The proposed portal contributes to this objective by identifying unnecessary electricity consumption and helping organizations improve the efficiency of existing energy use.

The system does not attempt to increase energy production. Instead, it focuses on reducing avoidable consumption.

The relationship can be represented as:

Energy Consumption → Waste Detection → Corrective Action → Reduced Avoidable Consumption → Improved Energy Efficiency

Sustainable Development Goal 11

The proposed system also supports **SDG 11: Sustainable Cities and Communities**.

Buildings are major components of urban infrastructure. Improving the energy performance of educational institutions, offices, hospitals and other buildings contributes to more sustainable communities.

Although the initial prototype is intended for a single building, it can be expanded to monitor multiple buildings across a campus or organization.

Sustainable Development Goal 12

The solution is also related to **SDG 12: Responsible Consumption and Production**.

Electricity is an important resource required for building operation. Unnecessary consumption represents inefficient resource utilization. By identifying situations where electricity may be consumed without a corresponding need, the portal encourages responsible use of energy.

ISO 14001

ISO 14001 provides a framework for Environmental Management Systems. The proposed portal can support several activities associated with environmental management.

The system can help organizations:

- identify electricity consumption as an environmental aspect;
- monitor energy-related performance;
- identify recurring energy-waste situations;
- document corrective actions;
- compare performance over time.

For example, if a particular classroom repeatedly produces after-hours energy-waste alerts, the building administrator can investigate the cause and introduce a corrective procedure.

The portal does not itself provide ISO 14001 certification. It functions as a technological tool that can support environmental performance monitoring and continual improvement.

Life Cycle Assessment

Life Cycle Assessment evaluates environmental impacts throughout the life cycle of a product, process or system. For buildings, this includes construction, operation, maintenance and end-of-life stages.

The proposed system mainly addresses the **operational stage** of a building. During operation, electricity is consumed by HVAC systems, lighting, computers and other equipment.

Reducing unnecessary operational energy consumption can reduce the environmental impacts associated with electricity generation.

A complete LCA of a building is outside the scope of this project because it would require detailed information about construction materials, transportation, equipment manufacturing and other life-cycle stages.

PROPOSED ENGINEERING/IT SOLUTION

Overview

The proposed solution is an **Energy Waste Detection and Audit Portal**.

It is a responsive web application that analyses building usage information to identify potential energy-waste events. The system is designed as a **software-first solution**, meaning that physical sensors are not required for the initial prototype.

The portal can use manually entered or simulated data such as:

- room schedules;
- equipment operating times;
- estimated equipment power;
- building working hours;
- energy consumption.

In a future implementation, these inputs can be obtained automatically from smart meters and IoT devices.

The main objective is not simply to display energy consumption, but to answer:

Where and when is energy potentially being wasted?

Main Components

The proposed portal contains the following modules:

- 1. Building Data Input:** Allows users to enter information about rooms, schedules and electrical equipment.
- 2. Energy Usage Analysis:** Processes energy consumption and operating-duration data.
- 3. Schedule vs Usage Comparison:** Compares the expected operating period with actual equipment operation.
- 4. Energy Waste Detection:** Identifies potential waste using predefined rules.
- 5. Room-Level Energy Audit:** Provides detailed information about energy usage and detected problems in individual rooms.
- 6. What-If Energy Simulator:** Allows users to estimate potential savings by changing operating conditions.
- 7. Waste Report:** Summarizes detected events and identifies rooms or equipment requiring attention.

Energy Waste Detection

The main individual contribution of this proposal is the **Energy Waste Detection Module**.

The module uses simple rules to identify potentially unnecessary energy use.

For example:

Rule 1 – Operation Beyond Schedule

If:

Actual operating duration > Scheduled operating duration

the system generates a potential waste alert.

Rule 2 – After-Hours Operation

If equipment operates outside predefined building working hours, the system flags it for investigation.

Rule 3 – Excessive Operation

If equipment operates significantly longer than the expected duration, the system identifies a potential energy-waste event.

These rules allow the system to detect problems without requiring complex artificial intelligence.

Example

Consider a classroom with the following information:

Parameter	Value
Room	204
Scheduled usage	9:00–10:00 AM
AC operation	9:00 AM–12:00 PM
Estimated AC power	2 kW

The scheduled operating duration is: **1 hour**

The actual operating duration is: **3 hours**

Therefore:

Excess operation = 3 – 1 = 2 hours

Potential excess energy can be estimated using:

$$E = P \times t$$

where E is energy in kWh, P is power in kW and t is time in hours.

Therefore:

$$E = 2 \times 2 = 4 \text{ kWh}$$

The system can display:

Potential Energy Waste Detected

Room 204 operated approximately 2 hours beyond its scheduled period.

Estimated potentially avoidable energy: **4 kWh**.

This provides the administrator with information that can be investigated and acted upon.

What-If Energy Simulator

The portal can also include an interactive simulator.

For example, suppose 10 rooms each have an estimated unnecessary AC operation of 2 kW for 1 hour per day.

Potential daily saving:

$$10 \times 2 \times 1 = 20 \text{ kWh/day}$$

For 30 days:

$$20 \times 30 = 600 \text{ kWh/month}$$

The user can modify values such as:

- number of rooms;
- equipment power;
- unnecessary operating time.

The portal then calculates the potential energy-saving opportunity.

The calculated values are estimates and depend on the accuracy of the input data.

Proposed Website

The prototype can be implemented using **HTML, CSS and JavaScript**.

The main webpages can be:

Dashboard

Displays:

- rooms monitored;
- total detected waste events;
- high-priority events;
- estimated avoidable energy;
- potential savings.

Room Analysis

Shows the schedule, equipment operation and detected issues for a selected room.

Waste Audit

Lists detected energy-waste events and their severity.

What-If Simulator

Allows users to modify operating conditions and calculate potential savings.

These pages satisfy the requirement for multiple functional webpages while keeping the system practical to implement.

FEASIBILITY ANALYSIS

Technical Feasibility

The proposed system is technically feasible because the prototype can be developed using standard web technologies.

HTML can be used for webpage structure, CSS for responsive design and JavaScript for calculations and waste-detection logic.

The prototype does not require hardware because building information can be simulated or entered manually.

A future version can connect the software to smart meters, IoT devices or existing Building Management Systems.

Economic Feasibility

The initial prototype has minimal development cost because HTML, CSS and JavaScript are freely available technologies.

A real-world deployment would require additional infrastructure such as sensors, smart meters, servers and networking. However, the software prototype allows the concept to be tested before such investments are made.

Operational Feasibility

The system is intended for facility managers, building administrators and maintenance personnel.

The portal presents complex information in a simple form. For example, instead of displaying only raw energy readings, it can state:

Room 204: Equipment operated 2 hours beyond scheduled usage.

This makes the information easier to understand and act upon.

Scalability

The system can initially be deployed for a single building and later expanded to:

Room → Floor → Building → Campus → Multiple Buildings

The software architecture can also be extended to support real-time data collection in the future.

EXPECTED ENVIRONMENTAL BENEFITS

The major environmental benefit of the proposed system is the reduction of potentially avoidable electricity consumption.

Reduction in Energy Waste

By identifying equipment operating beyond required periods, the system can help administrators reduce unnecessary electricity consumption.

Reduction in Associated Emissions

Electricity generation can produce greenhouse-gas emissions depending on the energy source. Therefore, reducing unnecessary electricity consumption can also reduce associated environmental impacts.

Improved Building Efficiency

The system helps identify recurring inefficient practices and provides information for corrective action.

Better Resource Utilization

The objective is not to reduce essential electricity consumption but to identify consumption that may not be necessary.

Increased Environmental Awareness

Visual dashboards, reports and energy-saving simulations can help building users understand the relationship between equipment operation and energy consumption.

The actual environmental benefit will depend on whether the detected issues are investigated and corrective actions are implemented.

JUSTIFICATION OF THE PROPOSED SOLUTION

The proposed Energy Waste Detection and Audit Portal is suitable for the problem of energy-inefficient buildings because it focuses on a specific and measurable source of inefficiency: **unnecessary operational energy consumption**.

A conventional energy dashboard answers:

How much energy was consumed?

The proposed system attempts to answer:

Where and when was energy potentially wasted?

This makes the information more actionable for building administrators.

The solution is also different from a general AI energy-analysis system. Rather than attempting to provide broad AI-based recommendations, this proposal concentrates on detecting specific energy-waste events using building schedules, operating periods and energy data.

Another advantage is that the solution can be implemented quickly without hardware. This makes it suitable for a web-based academic prototype while still allowing future integration with IoT systems.

The **What-If Simulator** further improves the usefulness of the system by allowing users to explore possible savings before implementing changes.

For example, an administrator could compare:

- current equipment operation;
- one-hour reduction in unnecessary operation;
- two-hour reduction;
- improved shutdown practices.

The resulting estimates can support decision-making.

Therefore, the proposed system combines **energy auditing, waste detection and savings estimation** in a simple software platform.

CONCLUSION

Energy inefficiency in buildings is not limited to inefficient equipment. Significant energy can also be consumed unnecessarily when electrical systems operate beyond required periods or remain active when rooms and facilities are not being used.

This report proposed an **Energy Waste Detection and Audit Portal** as a software-based solution for identifying such situations. The system compares expected building operation with actual or reported equipment usage and identifies potential energy-waste events.

The key individual contribution is the **Energy Waste Detection Module**, which uses rules such as schedule comparison, after-hours detection and excessive operating-duration detection. The system can also estimate potentially avoidable energy using equipment power and operating time.

The proposed portal can be implemented using HTML, CSS and JavaScript without requiring physical hardware for the initial prototype. This makes it practical for rapid development while allowing future integration with smart meters, IoT devices and building management systems.

The solution supports SDG 7 through improved energy efficiency and also contributes to SDG 11 and SDG 12. It can support environmental-management activities associated with ISO 14001 and primarily addresses the operational stage of a building from an LCA perspective.

Overall, the Energy Waste Detection and Audit Portal provides a simple, scalable and actionable approach to improving building energy efficiency by helping organizations identify **where, when and how energy may be unnecessarily consumed**.

REFERENCES

[1] United Nations, “Ensure Access to Affordable, Reliable, Sustainable and Modern Energy,” *United Nations Sustainable Development*, 2025.

<https://www.un.org/sustainabledevelopment/energy>

[2] United Nations, “Goal 11 | Make cities and human settlements inclusive, safe, resilient and sustainable,” *United Nations*, 2015.

<https://sdgs.un.org/goals/goal11>

[3] United Nations, “Goal 12 | Ensure sustainable consumption and production patterns,” *United Nations*, 2025.

<https://sdgs.un.org/goals/goal12>

[4] O. for, “ISO 14001,” *ISO*, 2026. <https://www.iso.org/standard/14001> (accessed Sept. 07, 2026).

[5] Solar Impulse, “Solar Impulse Foundation Topic How to shape the future of construction?,” *@solarimpulse*, 2022.

<https://solarimpulse.com/topics/green-building>